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Two-dimensional V_2C@Se (MXene) composite cathode material for high-performance rechargeable aluminum batteries

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ABSTRACT

Compared to lithium-ion batteries, rechargeable aluminum batteries as the potential safer, cheaper, and higher capacity energy storage devices have received more and more attention and research. Herein, we successfully synthesized vanadium carbide (V_2C) through etching with lithium fluoride and concentrated hydrochloric acid solution. The composite two-dimensional layered structure (V_2C@Se) is formed after calcining with selenium. Through X-ray photoelectron spectroscopy and electrochemical analysis, it can be studied that the reversible redox reaction of V_2C@Se mainly includes V^{2+}/V^{3+}, V^{4+}/V^{5+}, and Se^{2-}/Se^{2+} in the charging-discharging process. By studying the density functional theory (DFT) calculation results, it can be found that after selenium treatment the adsorption and diffusion of $[AlCl_4]^-$ on V_2C surface are smoother than before (if functional groups such as -OH are considered). This method not only removed the -OH function group on the surfaces of V_2C layers to improve the reversible intercalation of $[AlCl_4]^-$, but also realized higher energy density through the combination of Se and V_2C layers. Due to the participation of selenium, the cathode materials can reversibly provide an initial discharge specific capacity of 402.5 mAh g^{-1} at 1 A g^{-1}, and 119.8 mAh g^{-1} can be remained even after 1000 cycles. This study can inspire researchers to develop high energy density and high stability aluminum batteries (Al-batteries) by exploring the methods of synthesizing intercalation-type MXene composite cathode materials.

1. Introduction

Because of the abundant reserves in the earth's crust for aluminum element, low cost, safety, high volumetric and mass capacity (8040 mAh cm^{-3} and 2980 mAh g^{-1}), and many other advantages, Al-batteries are expected to replace lithium-ion batteries (LIBs) as the new generation of energy storage device [1]. In 2011, N. Jayaprakash of Cornell University reported recyclable Al-batteries with 1-ethyl-3-methylimidazolium chloride ([EMI*m*]Cl) as the electrolyte for the first time [2], which solved the hydrogen evolution reaction problems of water-based Al-batteries. In 2015, Professor Dai's team developed an ultra-fast Al-batteries with graphene as cathode which can deliver a maximum discharge specific capacity of 70 mAh g^{-1}. There is no capacity fading even after 7500 cycles [3]. Therefore, finding the appropriate cathode materials for Al-batteries has become the goal of many researchers, including transition metal oxides, sulfide, selenides, telluride, organics, organic salts, carbon materials, two-dimensional materials, and many other types [4–11]. About vanadium Al-batteries, Achim M. Diem reported a V_2O_5 cathode for Al-batteries delivered a specific discharge capacity of up to ~170 mAh g^{-1}. But after 500 cycles, it only remained 25 mAh g^{-1} [12]. For Al-Se battery, the obvious shortcomings are poor cycle performance and low retention rate at high current density. In 2020, Jiao's group discovered an Al-Se batteries presented a maximum original specific capacity of 450 mA g^{-1}. The discharge capacities of the batteries remain steady (about 400 circles) at 66 mAh g^{-1} at 1000 mA g^{-1} [13]. Therefore, how to solve the problems of poor stability and rapid capacity attenuation of aluminum selenium batteries has gradually become the focus of researchers.

In 2011, as a new two-dimensional material MXene entered people's vision. It has attracted much attention because of its graphene-like structure and excellent properties. MXene is mainly divided into carbides and nitrides [14], but both can be represented by $M_{n+1}X_nT_x$ (n = 1, 2, or 3), the thickness of the 2D sheets are related to the 'n' in MXene. M represents early transition metals, such as Ti, V, Zr, Mo, etc. X represents C or N. And T represents the surface group after etching, 'x' means there are different kinds of the groups in different ratios, including -O, -Cl, -OH, - F, etc. [15]. It not only has the advantages of high electron transmission efficiency of carbon materials but also has a good stability of metal salts, and thanks to the flexibility of the M elements, there are quite a few members (about 60 kinds) of the MXene family [16–25].

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The layered structure of MXene makes it possible to construct thin-film flexible batteries. Jia's group found that the interaction between Ti_3C_2 and the PAN substrate makes it more flexible without affecting the excellent conductivity of Ti_3C_2 [26]. The plentiful active sites on the two-dimensional surface are conducive to the smooth process of energy storage ion reversible intercalation. In general, MXene has good flexibility, good self-assembly characteristics, and corrosion resistance, which undoubtedly will be leading the future development of Al-batteries with high energy density and excellent reversibility [27]. Among all materials of Al-batteries cathode, MXene shows excellent cycle performance and high discharge specific capacity. Recently, Li's team used Ti_3C_2@CTAB-Se cathode material of Al-batteries to achieve 583.7 mAh g^{-1} discharge capacity at a current density of 100 mA g^{-1}, and there is still 132.6 mAh g^{-1} even after 400 cycles [28]. Lee's team reported a Fe_2CS_2 MXene as cathode material of Al-batteries. Its capacity increases from 288 mAh g^{-1} (Ti_2CO_2) to 642 mAh g^{-1} due to the charge redistribution [29]. Vahidmohammadi and his team reported a vanadium carbide (MXene) cathode for high-capacity Al-batteries. During the discharge process, Al^{3+} cations produced by the dissociation of $[Al_2Cl_7]^-$ anions, which react at the electrode/electrolyte interface intercalate into the MXene interlayer area. Meanwhile, $[AlCl_4]^-$ and metal aluminum react to produce $[Al_2Cl_7]^-$ on the anode. Its initial capacity is 922 mAh g^{-1}, but the specific capacity soon (about 100 cycles) decreased to less than 100 mAh g^{-1} at 100 mA g^{-1} [30]. So far, the capacity attenuation problem of MXene cathode in aluminum batteries is still the focus of researchers.

In this work, multilayer V_2C was prepared by etching V_2AlC with concentrated hydrochloric acid (HCl) and lithium fluoride (LiF). At 1A g^{-1}, it has 208.8 mAh g^{-1} initial discharge capacity but after 1000 cycles only 31.5 mAh g^{-1} left. Using V_2C multilayer structure as the skeleton, a more stable two-dimensional V_2C@Se composite structure was formed by calcinating with selenium. The initial discharge specific capacity of V_2C@Se is 402.5 mAh g^{-1} at 1A g^{-1}, and it is still more than 100 mAh g^{-1} after 1000 cycles. V_2C@Se cathode material demonstrates outstanding performances because of the stable multilayer structure after calcination as well as the intercalation-deintercalation reaction of selenium in the V_2C@Se interlayers. In addition, the strong interaction between $[AlCl_4]^-$ and functional groups like -OH on the surface of MXene made the $[AlCl_4]^-$ hardly to be deintercalated during charging, which also causes the $[AlCl_4]^-$ collapse and decompose into the electrolyte.

2. Results and discussion

As shown in Fig. 1(a), the preparation process of two-dimensional V_2C@Se multilayered powder is divided into two steps. In the first step, the Al atomic layer of V_2AlC was selectively etched to form the two-dimensional V_2C multilayered structure. In the second step, V_2C powder was calcined with selenium at a temperature higher than the melting point (217 °C) of Selenium. Under high temperature the dissolution and diffusion of selenium happened between the V_2C layers, meanwhile on the surface of V_2C the functional groups were reduced and a more regular interlayer spacing was formed. The BET result of V_2C@Se is reduced from 7.6 m^2 g^{-1} (V_2C) to 5.3 m^2 g^{-1}, which is shown in Fig. S1. It also proved the functional groups of V_2C were reduced. The XRD patterns of V_2AlC, V_2C, and V_2C@Se are shown in Fig. 1(b), which shows that the main peak of V_2C at about 5θ characterizes the V_2C (002) crystal surface. It proves that the etching result is excellent [30]. In the XRD pattern of V_2C@Se, the (002) peak of the crystal plane disappears, and the main peak of it characterizes the selenium element, which proves that selenium is uniformly doped on the two-dimensional surface among V_2C layers after calcining. The Rietveld analysis result of V_2C@Se are shown in Fig. S2 and Table S1. In addition, the V_2C@Se after chloroform etching was prepared to confirm that Se was doped between the V_2C layers. The detailed description is shown in the supporting information (Fig. S3). Fig. 1(c) and (d) are the SEM image of V_2AlC and multilayered V_2C particles, respectively. The expanded interlayer structure is not only conducive to the deintercalation reaction of $[AlCl_4]^-$, but also creates active sites for selenium doping. The SEM image of V_2C@Se is shown in Fig. 1(e). In the V_2C@Se structure, selenium is uniformly doped on the two-dimensional surface of V_2C layers, which is conducive to the stability of selenium in the deintercalation process and reduces the probability of selenium dissolution in the electrolyte during charging. The HRTEM images of V_2C and V_2C@Se are shown in Fig. 1(f, g) and (h, i), respectively. The layer spacing of V_2C and V_2C@Se is about 1.08 nm and 0.97 nm, respectively. After V_2C was doped by selenium, the layer spacing is reduced, but the lattice stripes are much clearer, which indicates that after calcination with selenium, the functional groups such as -O, -OH, -F, -Cl on the surface of V_2C are reduced and replaced by selenium atoms [28]. Moreover, the loss of irregular functional groups leads to a decrease in the layer spacing and regularity of the lattice structure, which is conducive to improve the cycle performance of Al-batteries.

Through the XPS analysis, the bonding and elements valence states changes before and after selenium treatment of V_2C cathode are characterized. As shown in Fig. 1(j), two of the V_2C XPS spectrum binding energy peaks at 513.5 eV and 521.4 eV are attributed to V^{2+} $2p_{3/2}$ and V^{2+} $2p_{1/2}$, respectively. At 516.5 eV and 524.2 eV there are two peaks belong to V^{4+} $2p_{3/2}$ and V^{4+} $2p_{1/2}$, respectively. The XPS spectrum of V 2p in V_2C@Se is shown in Fig. 1(l). Compared with Fig. 1(j), the content of V^{2+} decreased rapidly. Meanwhile, the content of V^{4+} did not change much. This shows that V^{2+} was oxidized to V^{4+} after selenization [30]. From Fig. 1(k) the XPS spectrum of C 1 s in V_2C, C-V bond is found to be the main component except for C–C bond at 284.8 eV. While the amount of C–O bond which is derived from the -OH functional group of H_2O is small, the content of O–C = O bond at 288.7 eV is even less [28]. Contrast the XPS spectrum of C 1 s in Fig. 1(k) and Fig. 1(m) C-V bond disappears after selenium treatment and the content of O–C = O bond increases, which indicates that a large amount of selenium doping destroys the C-V bond between the previous layers. At the same time, part of -OH is reduced to hydrogen gas and O–C = O bond through selenium treatment [31]. Without the influence of -OH, the cathode is more conducive to reversibly intercalate $[AlCl_4]^-$ anions, which will greatly improve the cycle performance.

The charging-discharging curves of the first two cycles and the fifth cycle of V_2C and V_2C@Se cathode materials are carried out to characterize their electrochemical performance. As shown in Fig. 2(a, b), Table S2, and S3 at a constant current density of 1 A g^{-1}, the initial discharge capacities of V_2C and V_2C@Se are 208.8 mAh g^{-1} and 402.5 mAh g^{-1}, respectively. And in the fifth cycle their discharge capacities are reduced to 77.1 mAh g^{-1} and 268.3 mAh g^{-1}, respectively. And in the fifth cycle their discharge capacities are reduced to 268.3 mAh g^{-1} and 77.1 mAh g^{-1}, respectively. The retention rate of V_2C cathode is only 36.94%. Obviously, compared with V_2C@Se the capacity fading of the V_2C cathode was too fast. That was because the function groups like -OH at the surface of V_2C layers were replaced by selenium after calcination, which prevented the collapse of $[AlCl_4]^-$ due to the strong interaction between -OH function groups and $[AlCl_4]^-$. Therefore, the problems of low discharge specific capacity and low coulombic efficiency are solved. The selenium treatment not only solved these problems, due to the stable structure of V_2C layers, the reversible intercalation process of $[AlCl_4]^-$ is also easier. This prevented the dissolution of selenium in the electrolyte and realized the goal of excellent cycle performance. The discharge curve in Fig. 2(a) tends to be flat near the cut-off voltage, which also confirms the collapse and dissolution of $[AlCl_4]^-$. Fig. 2(d) shows the charging-discharging curves of V_2C@Se at different current densities from 1 A g^{-1} to 3 A g^{-1}. The initial discharge capacity of V_2C@Se is 402.5 mAh g^{-1} at 1 A g^{-1}, 232.2 mAh g^{-1} at 1.2 A g^{-1}, 184.2 mAh g^{-1} at 1.5 A g^{-1}, 168.4 mAh g^{-1} at 1.8 A g^{-1}, 155.8 mAh g^{-1} at 2 A g^{-1}, 149.4 mAh g^{-1} at 2.5 A g^{-1}, and 120 mAh g^{-1} at 3 A g^{-1}. As shown in Fig. 2(e) after 35 cycles, the current density changes from 3 A g^{-1} to 1 A g^{-1}, V_2C@Se still has a stable discharge

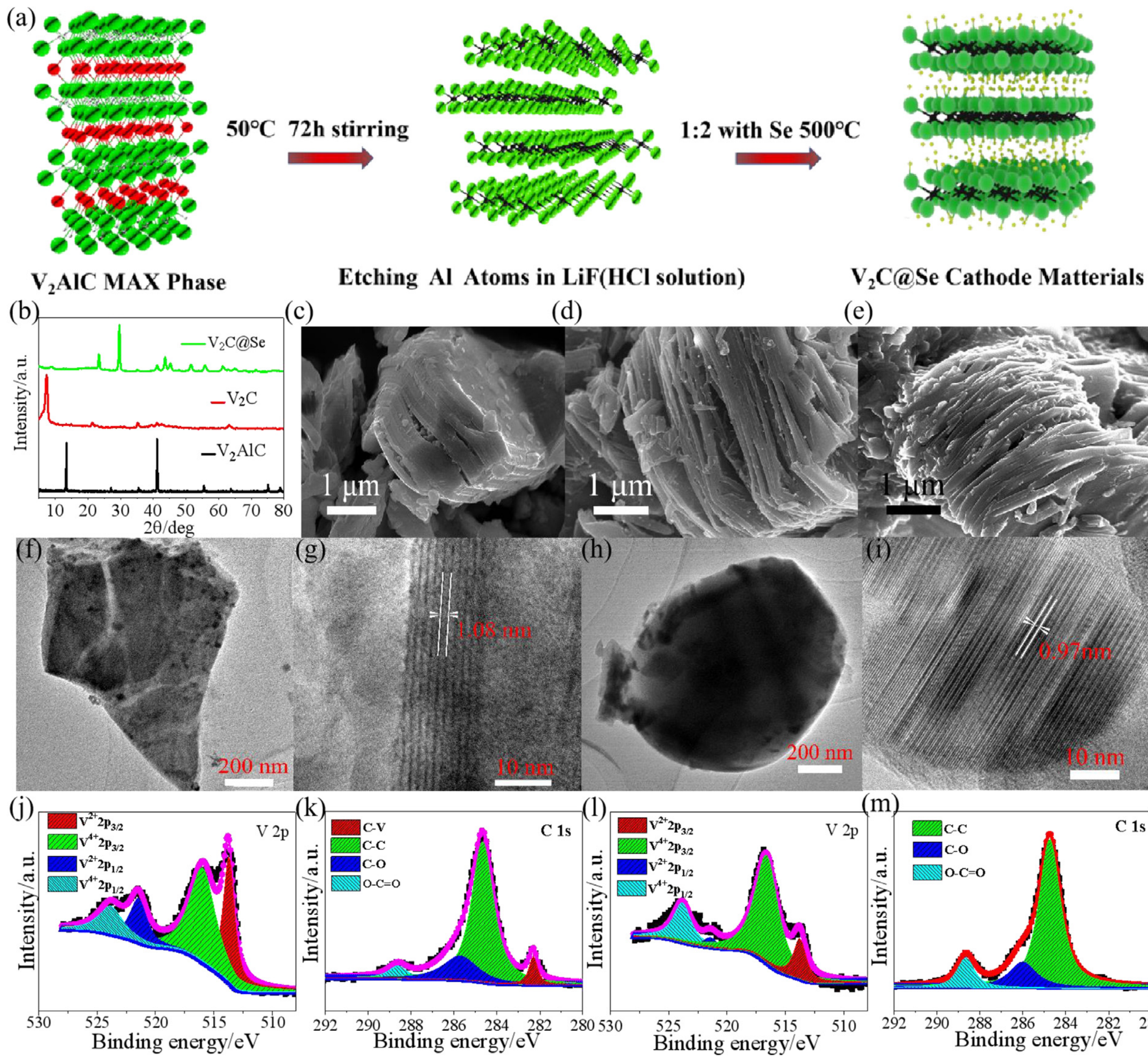

Fig. 1. (a) Diagrammatic sketch of V_2C@Se synthesis process. (b) XRD patterns of V_2AlC, V_2C, and V_2C@Se. SEM images of V_2AlC (c), V_2C (d), and V_2C@Se (e). TEM images of V_2C (f, g) and V_2C@Se (h, i). The V 2p and C 1 s region of V_2C (j, k) and V_2C@Se (l, m) XPS spectra.

specific capacity of 300.5 mAh g^{-1}, while the discharge specific capacity of V_2C has reduced to less than 100 mAh g^{-1}. This shows that V_2C@Se has excellent performance under high current density. Fig. 2(f) shows the cycle performance of V_2C@Se and V_2C at 1 A g^{-1}. It can be seen that after 1000 cycles, the discharge specific capacity of V_2C@Se is still 119.8 mAh g^{-1}, and the coulombic efficiency is till 92.65%. The SEM and EDS results of the V_2C@Se cathode (Fig. S4) prove that the V_2C structure is still complete and $[AlCl_4]^-$ can reversibly intercalate even after 1000 cycles. Meanwhile, the discharge specific capacity and coulombic efficiency of V_2C are only left 31.5 mAh g^{-1} and 76.33%, respectively. In contrast, after calcining with selenium, the cathode can reversibly intercalate/deintercalate $[AlCl_4]^-$ for energy storage even at high current density, which also helps to improve cycle performance and rate performance.

To characterize the kinetic characteristics of V_2C@Se, the cyclic voltammetry (CV) tests of V_2C, V_2C@Se, and pure molybdenum sheet are carried out, which is shown in Fig. 2(g). Obviously, the redox peaks of V_2C and V_2C@Se appear in different voltage but all have two pairs. Mo does not participate in the cathode reaction can be proved by the flat curve with no peaks. The first and second (reduction/oxidation) peaks of V_2C represented by A_1/A_2 and B_1/B_2 are about (1.60 V/1.70 V) and (1.75 V/2.20 V), respectively. For V_2C@Se the first and second reduction/oxidation peaks appear at (1.55 V/1.90 V) and (1.75 V/2.23 V), respectively. The redox peaks A_1/A_2 and B_1/B_2 are attributed to the redox reaction between V^{2+}/V^{3+} and V^{4+}/V^{5+}, respectively. The first three cycles CV curves of V_2C and V_2C@Se are shown in Fig. S5. To further understand the charging-discharging process of V_2C@Se, ex-situ XRD characterization analysis is carried out. As shown in Fig. 2(c), there are mainly three peaks (100), (101), and (102) that characterize the selenium crystal plane before the charging-discharging process. In the process of charging, the three peaks of selenium crystal planes decrease gradually. When it charged to the cut-off voltage of 2.4 V, the three peaks disappeared. After that, the three peaks increase gradually until it discharges to the cut-off voltage (0.1 V) in the discharge process. These peaks return to the initial state, which is consistent with the intercalation/deintercalation process of $[AlCl_4]^-$ in Al-batteries. In addition, we compared the electrochemical performances of V_2C@Se such as initial capacity, first circle current density, number of cycles, cycling current density, and after cycling discharge specific capacity with other cathode materials of Al-batteries in recent years. As shown in Fig. 2(h) and Table S4, the comprehensive electrochemical performance of V_2C@Se is outstanding [5, 7, 12, 32-47].

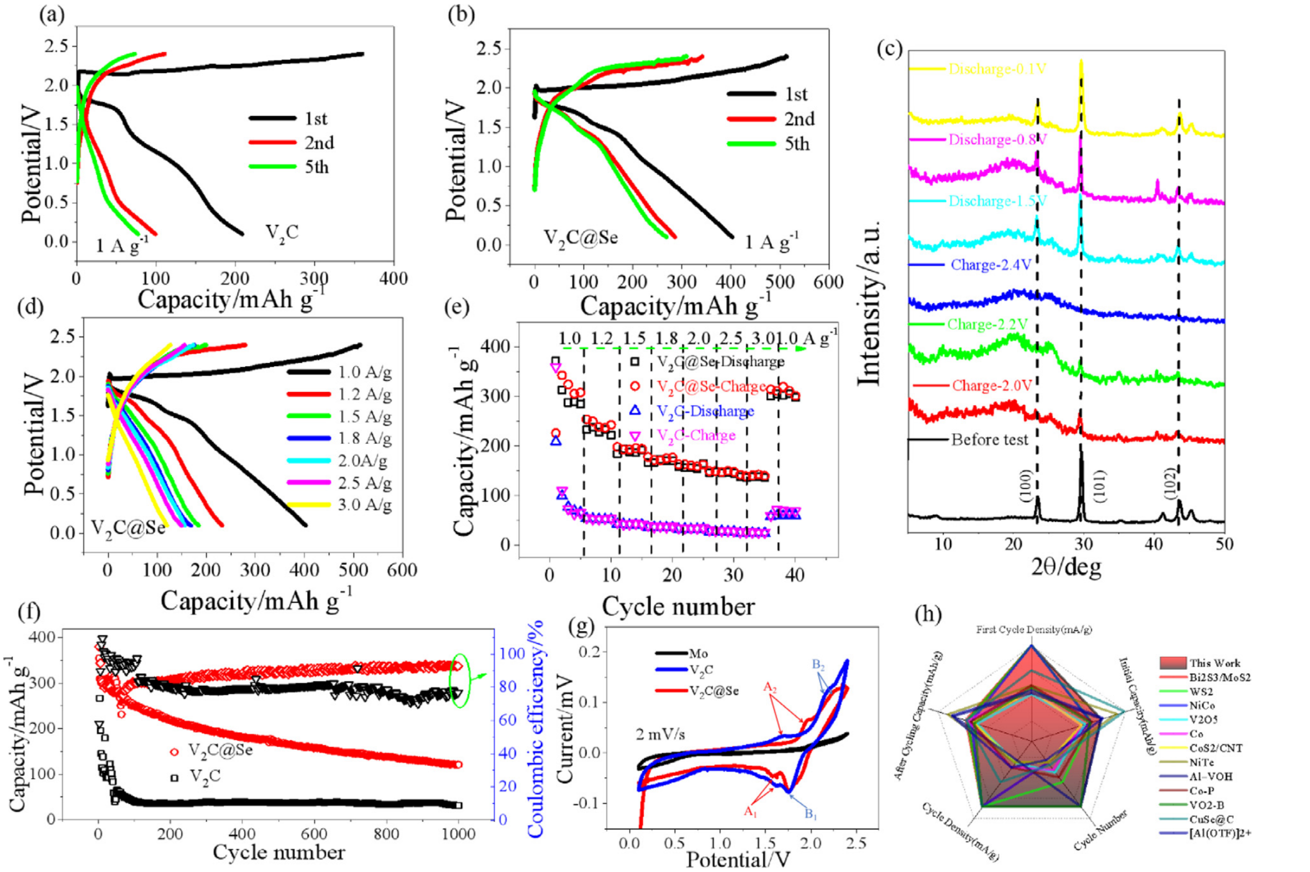

Fig. 2. Charging/discharging curves of V_2C (a) and $V_2C@Se$ (b) for the first, second, and fifth cycles. (c) Ex-situ XRD patterns of $V_2C@Se$ at different charging/discharging voltage. (d) Charging/discharging curves of the $V_2C@Se$ at different current densities. (e) Rate performance of V_2C and $V_2C@Se$ over 40 cycles at 1000 mA g^{-1} to 3000 mA g^{-1}. (f) Cyclic performance of V_2C and $V_2C@Se$ over 1000 cycles at 1000 mA g^{-1}. (g) The CV diagrams of Mo (collecting fluid), V_2C, and $V_2C@Se$ at 2 mV s^{-1}. (h) Comparison of electrochemical performance between this work and other cathode materials for aluminum batteries.

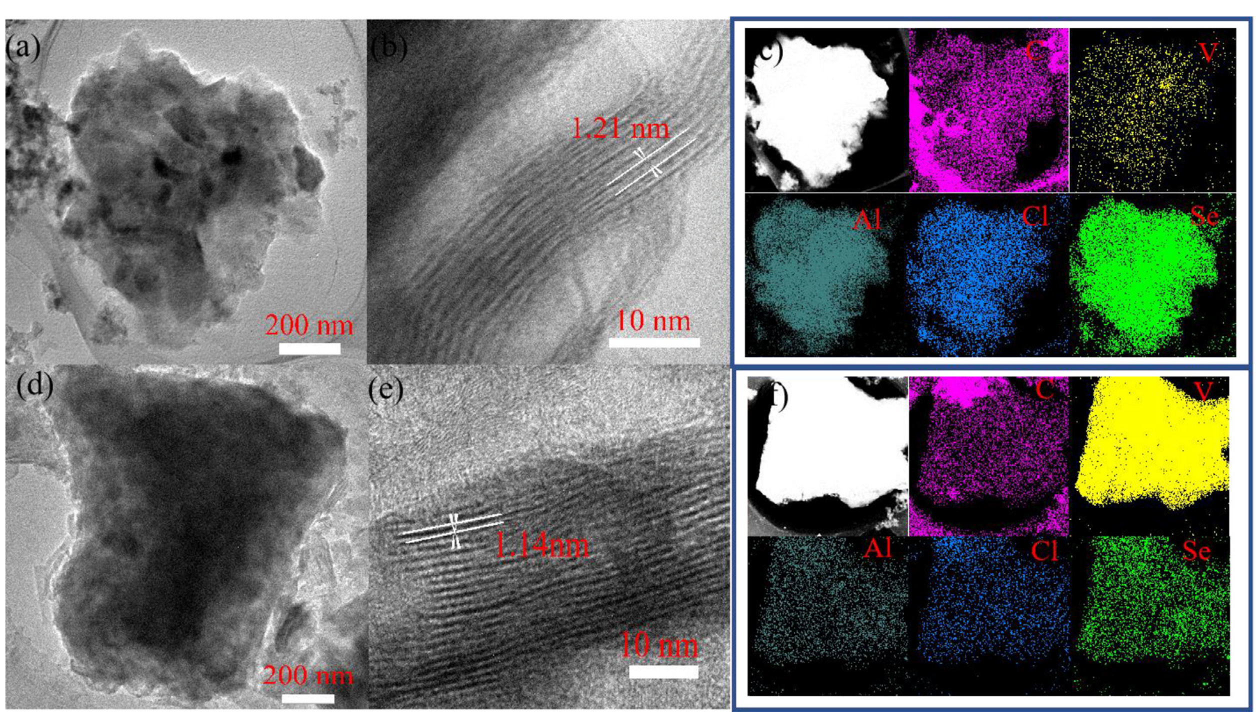

Fig. 3. TEM image of $V_2C@Se$ cathode charging to 2.4 V (a, b) and discharging to 0.1 V (d, e). Elemental mapping diagram of $V_2C@Se$ cathode charging to 2.4 V (c) and discharging to 0.1 V (f).

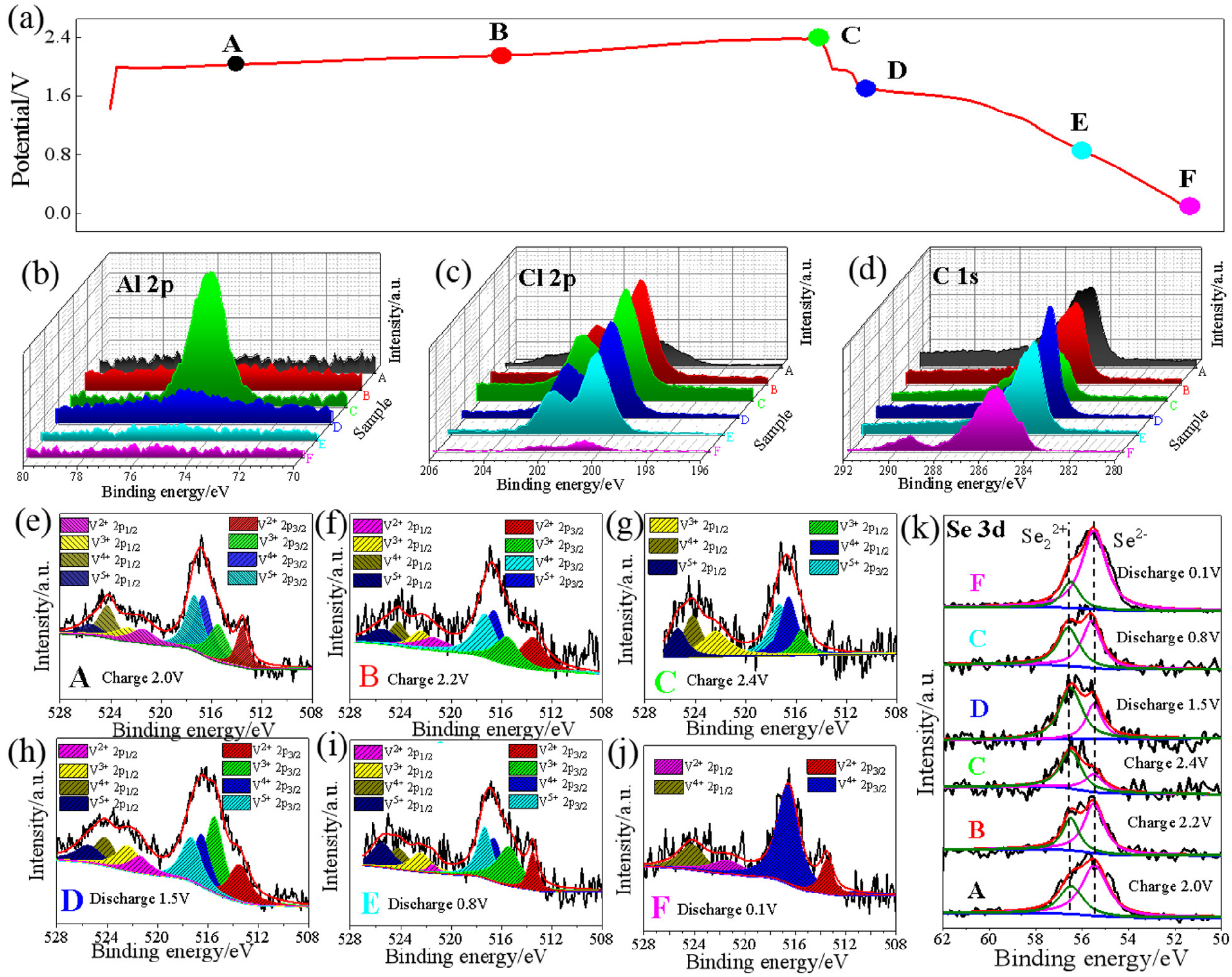

Fig. 4. Take A 2.0 V, B 2.2 V, C 2.4 V, D 1.5 V, E 0.8 V, and F 0.1 V six voltage points at V_2C@Se cathode charge-discharge line (a). XPS spectrum of Al 2p(b), Cl 2p(c), C 1s(d), and Se 3d(k) at these voltage points. XPS spectrum of V 2p at A(e), B(f), C(g), D(h), E(i), and F(j) six voltage points.

To figure out the relationship between the changes of structure and the behaviors of ions, the TEM analysis was carried out. Comparing the V_2C@Se TEM image of Fig. 3(b) with Fig. 3(e), when the V_2C@Se cathode is discharged from 2.4 V to 0.1 V, its layer spacing decreased from 1.21 nm to 1.14 nm. Comparing the elemental analysis maps of charging to 2.4 V (Fig. 3(c)) with that of discharging to 0.1 V (Fig. 3(f)), it can be found that the contents of aluminum, chlorine, and selenium are decreased after discharging process. This shows that a large number of $[AlCl_4]^-$ are deintercalated during the discharging process, which is consistent with the results of our theoretical calculation. This also explains the decrease in layer spacing [48]. But the ratio of aluminum to chlorine is more than 1:4, which can be explained by the irreversible adsorption of Al^{3+} and the decomposition of electrolyte [49–50]. The V_2C@Se cathode mainly reversibly intercalates $[AlCl_4]^-$ for energy storage. Without the strong interaction with the host lattice caused by the high charge density like Al^{3+}, even at high current density V_2C@Se can still provide an excellent cycle performance. The reduction of selenium may cause by its dissolution in the electrolyte and unable to be reversibly intercalated by the V_2C@Se interlayer during the discharging process.

The mechanism of electrochemical energy storage is further studied through ex-situ XPS. Different cut-off voltages (A: Charging to 2.0 V, B: Charging to 2.2 V, C: Charging to 2.4 V, D: Discharging to 1.5 V, E: Discharging to 0.8 V, and F: Discharging to 0.1 V) are shown in the charge and discharge curves, as shown in Fig. 4(a). Fig. 4(b), Fig. 4(c), and Fig. 4(d) show the XPS of Al 2p, Cl 2p, and C 1 s under different charging and discharging states. Obviously, the content of Al 2p and Cl 2p tends to increase which is caused by the intercalation of $[AlCl_4]^-$ during the charging process, on the contrary, due to the release of $[AlCl_4]^-$ during the discharge process, the content of Al 2p and Cl 2p tends to decrease. The XPS peak of C 1 s has basically not changed, which proves that it is not directly involved in the electrochemical reaction. Compared to the initial state (Fig. 2(l)), when charged to 2.0 V (Fig. 4(e)), the new peaks observed at 515.4 eV, 517.5 eV, 522.4 eV, and 525.7 eV are attributed to V^{3+} $2p_{3/2}$, V^{5+} $2p_{3/2}$, V^{3+} $2p_{1/2}$, and V^{5+} $2p_{1/2}$ respectively. This advantageously shows that the intercalation of $[AlCl_4]^-$ makes V^{2+} and V^{4+} oxidized to V^{3+} and V^{5+}, respectively. When charged to 2.2 V (Fig. 4(f)), due to the further intercalation reaction of $[AlCl_4]^-$, the content of V^{2+} and V^{4+} decreased, while the content of V^{3+} and V^{5+} were gradually increasing. When the cathode was completely charged (Fig. 4(g)), V^{2+} was basically not found, indicating that the transition to a high-valence state has been completed. When the next step was to discharge to 1.5 V (Fig. 4(h)), it is obvious that V^{2+} reappeared, which strongly indicates that due to the release of $[AlCl_4]^-$, the high-valence state of V changed to the low-valence state. During the discharging process, the XPS peaks of V^{2+} and V^{4+} gradually increased, while the XPS peaks of V^{3+} and V^{5+} had a trend of gradual decrease (Fig. 4(i)). When the discharge is completed (Fig. 4(j)), the XPS peak basically returned to the original state, which explains the reversible reaction between V^{2+}/V^{3+} and V^{4+}/V^{5+}. According to Fig. 4(k), the redox reactions of selenium element were between Se^{2-}/Se^{2+}. During charging process (from point A to point C) the content of Se^{2+} was increasing which is relate to the in-

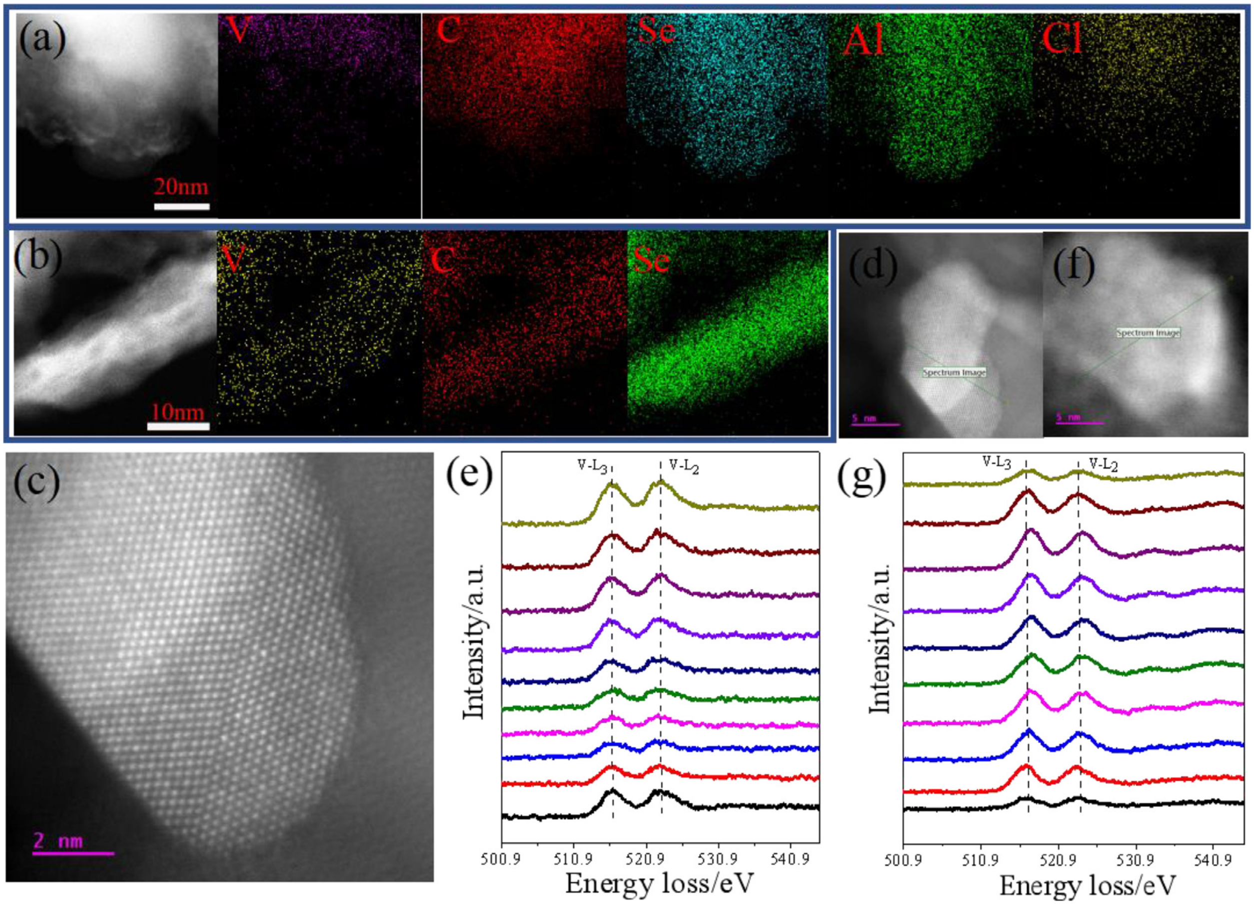

Fig. 5. (a) The low-magnification STEM image and EDS mapping of V_2C@Se cathode after 1000 cycles for V, C, Se, Al, and Cl elements. (b) The low-magnification STEM image and EDS mapping of V_2C@Se for V, C, and Se elements. (c) The HRTEM image of V_2C@Se. (d) and (f) HAADF image of V_2C@Se for EELS line scanning pathway as indicated by the green line. (e) EELS spectrum of V- L_3 and V-L_2 edges along the scanning pathway in (d). (g) EELS spectrum of V- L_3 and V-L_2 edges along the scanning pathway in (f).

tercalation of $[AlCl_4]^-$. Meanwhile during the discharging process (from point C to point E), due to the release of $[AlCl_4]^-$ the content of Se^{2-} was decreasing. When it was fully discharged (point F), the content of Se^{2+} backed to the initial level and the content of Se^{2-} went to the highest level. Therefore, the above analysis shows that the reversible redox reaction of V_2C@Se is mainly V^{2+}/V^{3+}, V^{4+}/V^{5+}, and Se^{2-}/Se^{2+}.

The EDS results in Fig. 5(a) proves that after 1000 cycles, selenium can still reversibly intercalate. The amount of aluminum and chlorine suggests that Al^{3+} and $[AlCl_4]^-$ are all participate in the charging/discharging process. The EDS results in Fig. 5(b) shows the distribution of V, C, and Se elements. As shown in Fig. 5(c), the HRTEM image of V_2C@Se has distinct layered lattice. These two results suggest that the selenium element is uniformly doped on the two-dimensional surface among V_2C layers. As shown in Fig. 5 (d and e), the result of EELS spectrum shows that the energy loss of V- L_3 and V-L_2 edges remains the same. But as shown in Fig. 5 (f and g), the result is different. The energy loss peaks of V- L_3 and V-L_2 edges are shift to the right, when it is scanned from the surface to the inside. But when it is scanned to another side, the peaks are shift back to the initial position, which suggests that the interlayer and outer surface energy storage mechanism of V_2C@Se multi-layer structure is different. This is consistent with the DFT results.

The adsorption energy (Eads) is calculated to quantitatively analyze the interaction between $[AlCl_4]^-$ species and substrates as:

$$Eads = E_{[AlCl4]-/Al} + E_{substrate} - E_{[AlCl4]-/Al+substrate}$$

where $E_{[AlCl4]\text{-}/Al+substrate}$, $E_{[AlCl4]\text{-}/Al}$, and $E_{substrate}$ represent the total energies of the adsorbed system, $[AlCl_4]^-$/Al, and substrates, respectively.

The density of states for V_2CO (Fig. 6(a)) and V_2CSe (Fig. 6(b)) show that around 0 eV their densities of states are all successive, which proves that they are all conductors. The adsorption energies of V_2CO and V_2CSe for $[AlCl_4]^-$ are −0.32 eV and −1.8 eV (Fig. 6(c)), respectively, which means that V_2CSe is easier than V_2CO to adsorb $[AlCl_4]^-$. The adsorption energies of V_2CO and V_2CSe for Al^{3+} are −1.6 eV and −2.32 eV (Fig. 6(d)), respectively. The diffusion paths of $[AlCl_4]^-$ (Fig. 6(e)) and Al^{3+} (Fig. 6(h)) on the surface of the V_2C structure show that the diffusion distance of $[AlCl_4]^-$ is smaller than Al^{3+}. The diffusion barriers calculation of V_2CO (Fig. 6(f)) and V_2CSe (Fig. 6(g)) for $[AlCl_4]^-$ are 0.12 eV and 0.21 eV, respectively. Although the diffusion barrier of V_2CO for $[AlCl_4]^-$ is 0.09 eV less than V_2CSe, the adsorption energy of V_2CO (−0.32 eV) is −1.48 eV less than V_2CSe (−1.8 eV), which shows that V_2CSe is easier to intercalate $[AlCl_4]^-$ than V_2CO. The diffusion barriers calculation of V_2CO (Fig. 6(i)) and V_2CSe (Fig. 6(j)) for Al^{3+} are 0.68 eV and 0.76 eV, respectively. In V_2CSe, the adsorption energy for Al^{3+} (−2.32 eV) is only 1.29 times the adsorption energy for $[AlCl_4]^-$ (−1.8 eV). But the diffusion barrier for Al^{3+} (0.76 eV) is 3.62 times the diffusion barrier for $[AlCl_4]^-$ (0.21 eV). It means that Al^{3+} is easier to be adsorbed than $[AlCl_4]^-$ on the surface of the multi-layer structure. But most of the Al^{3+} are irreversible, which means $[AlCl_4]^-$ is the main energy storage ion in V_2CSe. Therefore, through the above analysis, it can be known that the V_2CSe cathode is more conducive to reversibly intercalate $[AlCl_4]^-$ anions, which significantly improves the electrochemical performance of Al-batteries.

Fig. 7 shows the diagrammatic sketch of the V_2C@Se cathode discharge process, which is based on the above analysis. When the battery is charging, on the anode side $[AlCl_4]^-$ and metal aluminum are

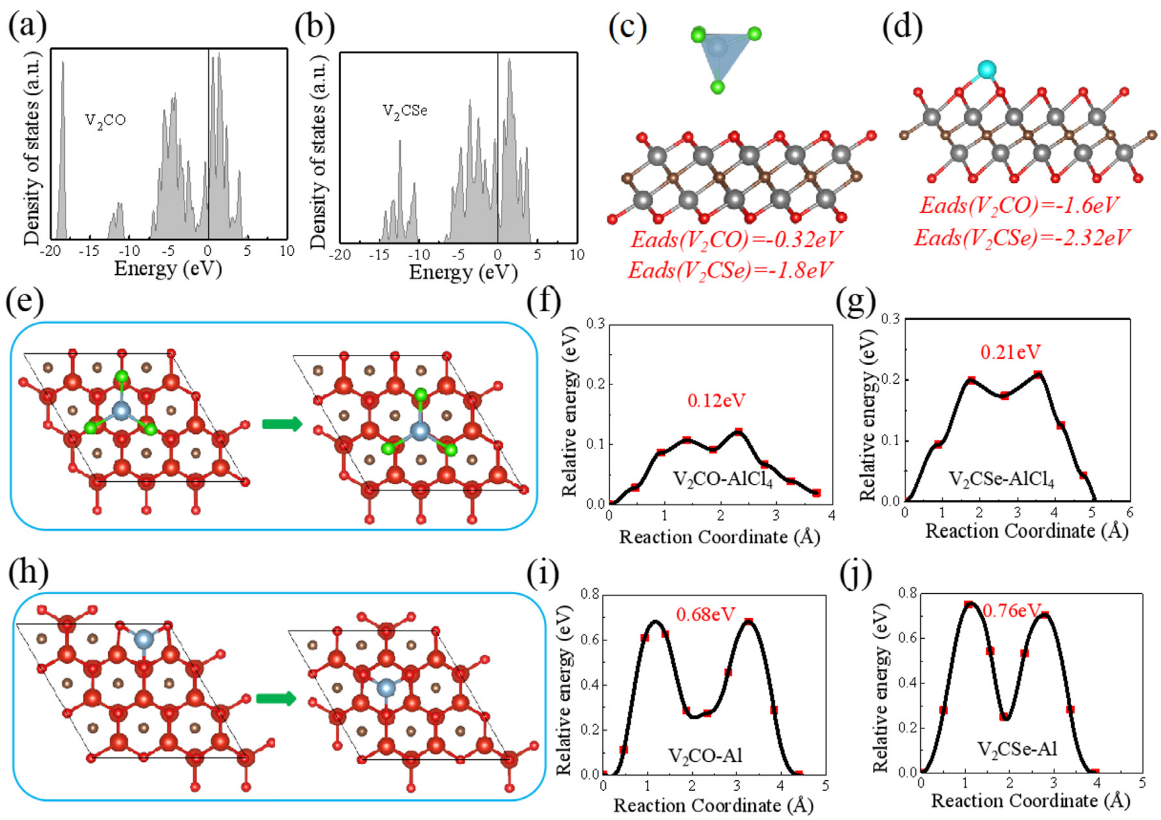

Fig. 6. The density of states for V_2CO(a) and V_2CSe(b). For $[AlCl_4]^-$ (c) and Al^{3+} (d), the adsorption energies of V_2CO and V_2CSe. The diffusion paths of $[AlCl_4]^-$(e). For $[AlCl_4]^-$, the diffusion barriers of V_2CO(f) and V_2CSe(g). The diffusion paths of Al^{3+}(h). For Al^{3+}, the diffusion barriers of V_2CO(i) and V_2CSe(j).

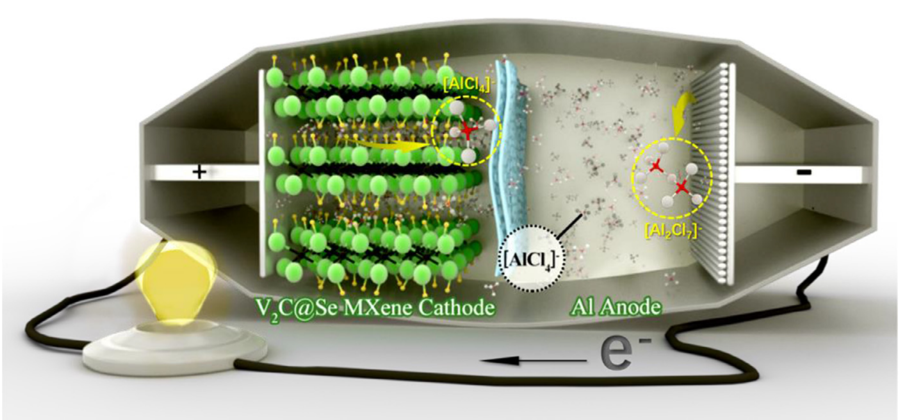

Fig. 7. The diagrammatic sketch of the $V_2C@Se$ cathode discharging process.

produced by the $[Al_2Cl_7]^-$ decomposition. Meanwhile the cathode is intercalating $[AlCl_4]^-$, which is the reason of the valence state of vanadium reversibly change from V^{2+} to V^{3+} or from V^{4+} to V^{5+} in $V_2C@Se$ cathode. When the battery is discharging, $[AlCl_4]^-$ and metal aluminum react to produce $[Al_2Cl_7]^-$ on the anode. Meanwhile the cathode is extracting $[AlCl_4]^-$ causes the reduction reaction of V^{3+} changes to V^{2+} or V^{5+} changes to V^{4+} in $V_2C@Se$ cathode. During the charge-discharge cycle, due to the intercalating-extracting process of $[AlCl_4]^-$ the valence state of selenium reversibly changes between Se^{2-} and Se^{2+}. According to the results above, the electrochemical reaction equations of $V_2C@Se$ cathode aluminum battery electrodes are established as follows:

Cathode:

$$V_2C + [AlCl_4]^- \Leftrightarrow V_2[AlCl_4]^-C$$

$$Se^{2-} + 4[AlCl_4]^- \Leftrightarrow SeCl_2 + 2[Al_2Cl_7]^-$$

Anode:

$$4[Al_2Cl_7]^- \Leftrightarrow 7[AlCl_4]^- + Al$$

3. Conclusion

We have successfully synthesized $V_2C@Se$ MXene composite cathode materials by selective etching aluminum element of V_2AlC through LiF and HCl solution then calcining with selenium. Through the selenium doping $V_2C@Se$ cathode provides excellent electrochemical performance. It is not only because of the improved surfaces of V_2C layers (without -OH) but also caused by selenium participating in the redox reaction (Se^{2-} and Se^{2+}) during the charging-discharging process. At 1 A g^{-1}, the $V_2C@Se$ cathode can provide a reversible maximum discharge specific capacity of 200 mAh g^{-1}. Benefit from the stable structure of V_2C layers, there is still 119.8 mAh g^{-1} after 1000 cycles. According to the DFT calculation results, after selenium treatment V_2CO is more conducive to the reversible intercalation of $[AlCl_4]^-$. Nevertheless, compare with similar works of Al-batteries, this work has great advantages, especially in high current density and excellent cycle performance. Further research can provide many kinds of application potential, such as in the vehicle industry field. However, there are always some of the selenium dissolved in the electrolyte. Further studies about how to reduce the dissolution of selenium elements will significantly improve the cycle performance of the composite cathodes like V_2C@Se.

Notes

The authors state that there are no competing economic interests.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Wenrong Lv: Writing – original draft, Data curation, Formal analysis. **Gaohong Wu:** Visualization, Investigation. **Xiaoxiao Li:** Visualization, Investigation. **Jianling Li:** Writing – review & editing, Supervision. **Zhanyu Li:** Conceptualization, Methodology, Writing – review & editing, Funding acquisition.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ensm.2022.01.019.

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